Assignment 4: Decision Tree Model

Business Interpretation Based on a Business Plan

Evergreen Bank — Credit Card Upgrade Program

Course	MBAI 5310G-001 — AI Programming
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Dataset	evergreen_bank_credit_card_upgrade_dataset.xlsx

Task 1: Understand the Business Problem

Business Overview

Evergreen Bank is a mid-sized retail banking institution that offers everyday banking, savings products, loans, and credit cards to individual and small business customers. The bank operates in a competitive financial services market where customers can easily compare credit card rewards, travel benefits, cash-back offers, annual fees, and digital banking experiences.

The bank has identified credit card upgrades as a strategic growth opportunity. Instead of acquiring completely new customers, the bank can use existing relationships and transaction patterns to identify current customers who may benefit from a higher-tier card. This approach is less expensive than broad acquisition campaigns because the bank already has customer history and communication channels.

Business Element	Description
Business name	Evergreen Bank Credit Card Services
Product focus	Premium credit card upgrades with stronger rewards, travel benefits, and customer support features
Target customers	Existing cardholders who may benefit from a higher card tier
Revenue sources	Annual card fees, interchange fees, higher card usage, and long-term customer retention
Main decision	Which customers should be prioritized for a credit card upgrade offer?
Main risk	Sending offers to unsuitable customers or missing customers who are likely to accept

What problem is the business trying to solve?

The bank currently uses simple rule-based targeting for upgrade campaigns, such as sending offers to customers with high monthly spending or long tenure. While these rules are easy to understand, they can miss more complex patterns. For example, a customer with moderate spending but high travel spending, strong digital engagement, and frequent offer email openings may be more likely to accept than a customer with high spending but low satisfaction and recent late payments.

Machine learning can help the bank combine multiple customer signals and predict the probability that a customer will accept an upgrade. The Decision Tree model is appropriate because it produces a structure that can be interpreted in business language.

Machine Learning Decision

Assignment Question	Answer for this Business Case
What is the target variable?	Accepted_Card_Upgrade
Possible target values?	Yes = the customer accepted the upgrade; No = the customer did not accept
What decision can ML support?	Prioritize customers for credit card upgrade outreach
Who may use the model?	Marketing analysts, CRM managers, branch advisors, and product managers
What should be excluded?	Customer_ID — it is only an identifier
Kind of model required?	Decision Tree classification model

Input Features

The input features span five groups:

- Customer profile: Age_Group, Income_Band, Employment_Status, Region, Customer_Segment
- Banking relationship: Customer_Tenure_Months, Current_Card_Tier
- Card usage: Avg_Monthly_Card_Spend, Travel_Spend_Share_Pct, Dining_Spend_Share_Pct, Reward_Points_Balance
- Engagement: Preferred_Channel, Digital_App_Logins_30D, Prior_Offer_Emails_Opened_6M, Branch_Advisor_Contact
- Risk and experience: Credit_Score_Band, Credit_Utilization_Pct, Late_Payments_12M, Customer_Satisfaction_Score

Why is this prediction useful?

The prediction is useful because it helps the bank focus its outreach on customers who are more likely to respond positively. This can improve campaign efficiency, reduce wasted communication, and increase the chance that customers receive offers that match their spending needs and financial profile.

The model can also help the bank choose different follow-up strategies. High-probability customers may receive a personalized email or mobile app offer. Medium-probability customers may receive educational information about rewards. Customers with risk indicators may require human review before any premium card offer is made.

Task 2: Prepare the Data

Dataset Overview

The dataset was loaded from the provided Excel file. It contains 340 rows and 21 columns, representing existing credit card customers with demographic, behavioral, engagement, risk, and satisfaction-related variables.

Inspection Results

Metric	Result
Rows x Columns	340 x 21
Missing values	0 in all columns
Duplicate rows	0
Target distribution	211 No (62%), 129 Yes (38%)
Numerical columns	Age, Income, Website_Visits, Time_on_Site, Previous_Purchases, etc.
Categorical columns	Gender, City, Membership_Level, Email_Opened, etc.

Data Cleaning

A copy of the original dataset was created to avoid modifying the raw data. Customer_ID was removed as it is only an identifier with no predictive value. No duplicate rows or missing values were found. The cleaned dataset retained 340 rows and 20 columns.

Feature Engineering

Categorical features were one-hot encoded using pd.get_dummies(). The target variable was label-encoded (No = 0, Yes = 1). After encoding, the feature matrix expanded from 19 features to 51 features. The data was split 80/20 into training (272 records) and testing (68 records) using stratified sampling to preserve class proportions.

Task 3: Train the Decision Tree Model

Introduction to Decision Trees

A Decision Tree is a machine learning model that makes predictions by learning simple decision rules from the data. The rules are organized like a tree, where each node asks a question about a feature and each branch represents the answer. The model keeps splitting the data until it reaches a final prediction at the leaf nodes.

Decision Trees are useful in business analytics because they are easy to understand and can show which features are most important. For Evergreen Bank, the tree can help explain whether high spending, credit score band, digital engagement, or prior offer engagement matter most in predicting upgrade acceptance.

How Decision Trees Make Predictions

Starting from the root node, the tree splits data based on the feature that best separates the classes. For example, the tree might first ask: Is the Credit_Utilization_Pct less than 44.5? Depending on the answer, it continues splitting — perhaps asking about Travel_Spend_Share_Pct or Avg_Monthly_Card_Spend — until it reaches a leaf node that gives a final prediction of Yes or No.

The deeper the tree grows, the more specific the rules become. However, a tree that is too deep memorizes the training data instead of learning generalizable patterns. This is called overfitting.

Model Training and Evaluation

The Decision Tree was trained without depth constraints to observe its default behavior. The tree grew to a depth of 10 with 48 leaves.

Metric	Result
Training Accuracy	100.0%
Testing Accuracy	63.2%
Gap	36.8 percentage points
Precision (Yes class)	51.9%
Recall (Yes class)	53.8%
F1-Score (Yes class)	52.8%
True Negatives (TN)	29 — correctly predicted No
False Positives (FP)	13 — predicted Yes, actually No
False Negatives (FN)	12 — predicted No, actually Yes
True Positives (TP)	14 — correctly predicted Yes

Controlling Decision Tree Complexity

To reduce overfitting, the tree was retrained with max_depth=4. This produced the following improvement:

Metric	Result
Training Accuracy (max_depth=4)	84.6%
Testing Accuracy (max_depth=4)	69.1%
Gap (max_depth=4)	15.4 percentage points

Limiting tree depth significantly improved generalization. The gap dropped from 36.8 to 15.4 percentage points and testing accuracy improved from 63.2% to 69.1%.

Top Feature Importances

Metric	Result
Credit_Utilization_Pct	26.3%
Travel_Spend_Share_Pct	10.9%
Dining_Spend_Share_Pct	9.5%
Avg_Monthly_Card_Spend	7.6%
Reward_Points_Balance	7.1%
Credit_Score_Band_Excellent	5.7%
Customer_Satisfaction_Score	5.5%
Customer_Tenure_Months	4.6%
Prior_Offer_Emails_Opened_6M	3.3%

Task 4: Business Interpretation of Model Results

How well did the model perform?

The Decision Tree achieved a testing accuracy of 63.2% on the 68-record test set. Precision for the Yes class is 51.9%, recall is 53.8%, and F1-score is 52.8%. The model provides some lift over random targeting but leaves room for improvement. When constrained to max_depth=4, testing accuracy improved to 69.1%, confirming that a simpler tree generalizes better to new customer data.

Overfitting Analysis

The unconstrained model achieved 100% training accuracy but only 63.2% testing accuracy — a gap of 36.8 percentage points. This is a clear sign of overfitting: the tree memorized every training record perfectly but cannot reliably classify new customers it has not seen before. Applying max_depth constraints is essential before using this model for any real campaign targeting.

Model Outcome and Business Meaning

Model Outcome	Business Meaning for Evergreen Bank
True Positive (14)	The model correctly identifies a customer who will accept. This supports effective and efficient targeting.
True Negative (29)	The model correctly identifies a customer who will not accept. This avoids unnecessary outreach and protects customer experience.
False Positive (13)	The model predicts acceptance, but the customer does not accept. This may waste marketing budget and annoy customers with irrelevant offers.
False Negative (12)	The model predicts non-acceptance, but the customer would have accepted. This causes missed revenue and missed cross-selling opportunities.
High training vs testing gap	Suggests overfitting. The tree learned the training data too closely and may not generalize well to new customers.

Which evaluation metric matters most?

The most important metric depends on the bank's business priority. As the business plan notes:

- If the bank wants to avoid sending irrelevant offers and protect customer experience, precision is the most important metric — it measures how many predicted acceptors actually accept.
- If the bank is running a growth-focused campaign and does not want to miss many suitable customers, recall is the most important metric.
- A balanced F1-score can be used when both concerns matter equally.

For this campaign, precision is especially important because Evergreen Bank operates in a relationship-driven banking context where customer trust and satisfaction are critical long-term assets. Sending irrelevant premium card offers can reduce satisfaction and signal that the bank does not understand the customer.

Feature Importance — Business Interpretation

Feature	Business Meaning
Credit_Utilization_Pct (26.3%)	The top predictor. Customers with moderate utilization are actively using their card without being overextended — prime upgrade candidates.
Travel_Spend_Share_Pct (10.9%)	Customers spending heavily on travel are natural candidates for premium travel-reward cards.
Dining_Spend_Share_Pct (9.5%)	Lifestyle spending on dining correlates with premium card appeal.
Avg_Monthly_Card_Spend (7.6%)	Higher overall card activity signals an engaged cardholder who would benefit from premium rewards.
Reward_Points_Balance (7.1%)	Customers actively accumulating points already value the rewards program — likely ready for a higher tier.
Credit_Score_Band_Excellent (5.7%)	Excellent credit customers are eligible for premium products and represent lower risk.
Customer_Satisfaction_Score (5.5%)	Satisfied customers have a positive relationship with the bank and are more likely to respond favorably.
Customer_Tenure_Months (4.6%)	Long-standing customers have higher trust in the bank and are more likely to accept cross-sell offers.
Prior_Offer_Emails_Opened_6M (3.3%)	Customers who open offer emails are already engaged with bank communications — upgrade offers have a higher chance of being read.

How the Bank Can Use the Model Responsibly

In a real business setting, the model would not automatically send offers. Instead, it would support a prioritization list for marketing and customer relationship teams. Customers with high predicted acceptance may be reviewed for eligibility, then placed into an appropriate outreach journey based on preferred communication channel and customer profile.

The bank should combine the model result with:

- Product eligibility rules (minimum credit score, income requirements)
- Affordability checks to ensure the premium card suits the customer's financial profile
- Customer consent and privacy compliance
- A human review process before contacting customers with borderline scores or risk indicators

Limitation and Bias

The dataset is synthetic and contains only 340 records — too small for a reliable production model and a major contributor to overfitting. More critically, features like Income_Band, Age_Group, Employment_Status, and Region may create unequal targeting patterns if used carelessly. A model may appear accurate while still producing outcomes that disadvantage certain groups. For example, lower-income or younger customers may be systematically scored as unlikely to accept, even if individual customers within those groups genuinely qualify and would benefit. This kind of disparate impact must be evaluated before any real-world deployment in financial services, where regulatory obligations around fair lending and equal access apply.

Limitations of Tree-Based Models

- Overfitting: Without constraints, Decision Trees grow too deep and memorize training data, leading to poor performance on new, unseen customers.
- Instability: Small changes in training data can produce a completely different tree structure, making the model unreliable across different datasets.
- Imbalanced classes: The model may favor predicting the majority class (No) more often, which can reduce recall for the minority class (Yes).
- Limited generalization: A Decision Tree trained on 340 synthetic records cannot capture the full complexity of real banking customer behavior.
- No probability calibration: The model predicts a class label but does not always produce well-calibrated probability scores, limiting its use in risk-scored campaign prioritization.

Responsible AI Reflection

Machine learning models used in financial services can affect real customers and must be developed and applied responsibly. As the business plan states, model outputs should support human judgment rather than replace it.

- Fairness: Income band, age group, employment status, and region may create unequal targeting patterns. A model may appear accurate while still producing outcomes that disadvantage certain groups. Business teams must audit the model for disparate impact before deployment.
- Transparency: The Decision Tree is interpretable — managers and auditors can follow the decision rules. This is an advantage in banking where explainability supports regulatory compliance and customer trust.
- Human oversight: Customer needs, affordability, credit suitability, and regulatory compliance cannot be fully captured by a simple Decision Tree. Human reviewers must validate model outputs before campaigns go live.
- Data quality: This dataset is synthetic and simplified for educational purposes. Real banking data would include more detailed transaction history, regulatory constraints, product eligibility rules, privacy requirements, and customer consent considerations.

Final Conclusion

This report documented the full process of building and evaluating a Decision Tree classification model to support Evergreen Bank's credit card upgrade campaign.

The unconstrained model achieved 100% training accuracy but only 63.2% testing accuracy, showing severe overfitting with a 36.8 percentage point gap. When depth was limited to max_depth=4, testing accuracy improved to 69.1% with a much smaller gap of 15.4 points. The top predictors — Credit_Utilization_Pct, Travel_Spend_Share_Pct, and Dining_Spend_Share_Pct — reflect spending behavior and lifestyle fit, which are strong signals of upgrade acceptance.

From a business perspective, precision is the priority metric for this campaign, as sending irrelevant offers damages customer trust in a relationship-driven banking context. The 13 false positives represent wasted outreach and potential customer dissatisfaction, while the 12 false negatives represent missed revenue and cross-selling opportunities.

The model provides a useful decision-support tool but must be combined with product eligibility rules, affordability checks, customer consent, and human review before any upgrade offers are sent. A more robust deployment would benefit from a larger real-world dataset, hyperparameter tuning, and a more powerful ensemble algorithm such as Random Forest.